

Snooze Spiral: A Visualization for Cyclical Sleep Tracking

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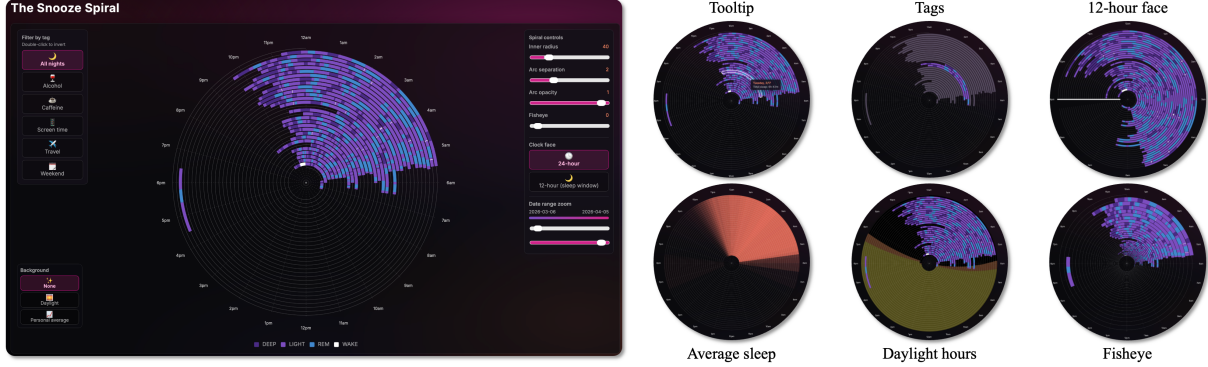


Figure 1: Visualization demo for a cyclical sleep visualization showing sample data and the effects of different interactive tools.

ABSTRACT

Around half of American adults track their sleep data, yet most sleep visualizations focus on daily sleep data. Visualizations of long-term sleep trends are primitive and may obscure actionable information from users. This contribution asks whether a cyclical approach to sleep visualization could improve people’s intuition about their sleep habits and patterns. A formative study evaluates how sleep trackers are used by young adults and a cyclical visualization is proposed as a solution to some of their challenges. The visualization is validated through a think-aloud study, and results indicate that while cyclical encodings improve pattern recognition and long-term reflection, the transition from descriptive insights to prescriptive behavioral change remains a persistent hurdle.

Index Terms: Sleep, cyclical, applications.

1 INTRODUCTION

Approximately half of American adults track their sleep [22], and more than half of them have changed their sleep habits as a result of tracking data [21]. About one-third of the population is struggling with their sleep [8]. Sleep is known to be of extraordinary importance to human health and flourishing [32]. Beyond consumer interests, there is also great interest in sleep patterns in many industries, from military personnel to air traffic control and healthcare [28]. Often, these industries employ off-the-shelf consumer sleep trackers (CSTs) to gather and analyze sleep data at an individual level.

As Van den Bulck notes, the proliferation of sleep apps has turned sleep into a measurable performance metric [27]. While this movement has democratized access to sleep science, it has also created a landscape where millions of users are drowning in data without the means to interpret it. Tools like SleepExplorer [19] demonstrate the sheer volume of metrics available - from sleep effi-

ciency to minutes awake - yet the availability of data has outpaced the development of intuitive ways to visualize it.

The overarching goal of this work is to contribute to the shallow landscape of sleep visualization techniques, especially those that help users gain insights over medium-term (weekly and monthly) timescales. These kinds of monthly sleep pattern visualizations are where the landscape of sleep visualizations is most barren, yet there is much to be learned from medium-term trends. We address this goal through four research questions:

RQ1: What challenges do sleep tracking app users face when interacting with in-app data visualizations?

RQ2: What variables do users track in conjunction with their sleep?

RQ3: Can visualizing sleep cyclically as opposed to linearly improve people’s intuition about their sleep habits and patterns?

RQ4: Does cyclical sleep visualization: (a) change what people notice/understand about their sleep, and (b) support better reflection than existing tools?

We address these research questions through two main contributions; first, a formative user study was conducted to understand how current users of sleep tracking devices interact with their data, addressing RQ1 and RQ2. Second, a cyclical sleep visualization technique is proposed and evaluated using a second user study, addressing RQ3 and RQ4.

2 RELATED WORK

To position this work in the field, related works are evaluated in the context of sleep data visualizations (Sec. 2.1) and in the context of cyclical encoding of time-series data (Sec. 2.2). This work brings the two fields together to produce a domain-specific visualization using a cyclical encoding.

2.1 Sleep Data Visualizations

A significant body of work has surveyed the current state of consumer sleep trackers (CSTs). Reviews by Choi et al. [7] and Ong & Gillespie [23] analyzed dozens of apps on the Google Play and Apple App Stores, finding that while 100% of apps report core metrics like sleep duration, the presentation of this data is remarkably homogenous. These surveys, along with user studies by Aravind [3]

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and Islam [15], confirm that the dominant visualization paradigms are linear: hypnograms, bar charts, and line graphs. This work challenges that paradigm by using a cyclical encoding.

Choi et al. [7] highlighted that few apps meet criteria for quality or functionality, often functioning as passive monitors rather than active self-management tools. Aravind [3] found that users primarily check “previous night” data and weekly overviews, preferring simple, familiar formats like bar charts even if they lack depth. Islam et al. [15] reinforced this, noting that users are most interested in short-term data (last night’s sleep) but often struggle to connect these daily snapshots to longer-term trends. These works highlight the challenge of revealing and highlighting long-term sleep trends and changes.

While surveys tell us what features exist, broader design research highlights a gap between what apps provide and what users actually need. Work by Tsai, Oyebode, and Aji shifts the focus to user pain points and the desire for actionable insight:

- Tsai [26] identified critical “pain points” in Fitbit users, specifically the lack of a clear baseline (“what is normal for me?”) and the absence of actionable advice.
- Oyebode [25] and Aji [1] emphasize that users are not just looking for data, but for personalization. Aji’s work on “diverse sleep phenotypes” suggests that a one-size-fits-all visualization fails to account for unique and irregular schedules.
- Ghosh [12], in the context of the VANESSA system, demonstrates that while more sophisticated tools are available to researchers, comprehensive visual analytics are not yet available for everyday users to interpret their sleep metrics and patterns.

Current tools effectively answer the question, “How did I sleep last night?” but fail to answer, “What is the shape of my sleep?” By adhering to linear paradigms, existing apps obscure the cyclical nature of sleep - phase shifts, regularity, and social jetlag. While the utility of a cyclical view has been demonstrated in specific use cases, such as tracking infant sleep patterns. These examples have historically been limited to binary data (asleep vs. awake). This work expands that concept into a tool for adult self-informatics, moving beyond basic duration tracking to include the complex, multivariate factors that drive adult sleep health.

These multivariate factors include not just sleep duration and stages (often categorized as ‘Deep’, ‘Light’, ‘REM’, and ‘Awake’), but a variety of other metrics that are measured and tracked. There is limited discussion in open literature about the range of metrics that are commonly available to users and there is little consistency in the measured metrics across CSTs. Modern CSTs use raw data from a range of sensors to calculate meaningful sleep metrics, although the accuracy of these metrics can vary [18, 8]. The raw data is often not made available to users or developers, and is instead reported in the form of simplified, post-processed metrics such as sleep duration (including wake/ sleep classification), and sleep stages (light, deep, REM) [8]. Sleep stages are challenging to accurately define, but are often one of the primary metrics in CST visualizations. Connections between sleep data and daytime activity are also valuable and collection of continuous data could be valuable for understanding sleep patterns, according to deZambotti et al. [8].

Therefore, we find that the state of the art in sleep visualization is limited to a set of homogeneous visual encodings that illustrate short-term data to novice users [23]. More sophisticated tools are restricted to expert domains [12], and few efforts have been made to provide users with a more comprehensive understanding of longer-term (i.e. beyond nightly) sleep trends [15]. This is despite the wealth of sleep data that users are able to track; demonstrating that sleep tracking data is not being used to its fullest extent.

2.2 Spiral Visualizations for Time-Series Data

Spiral visualizations are a popular and effective choice for representing time-series data, particularly when cyclical patterns are present. Researchers have shown that cyclical visualizations are often much more effective than linear ones for understanding periodic phenomena, such as sunshine intensity, where spirals make it easier to compare days and identify specific events like sunrise and sunset [31]. These visualizations have been applied across diverse domains, including music rhythm analysis to account for recursive temporal frameworks [4], the visualization of the ovulatory menstrual cycle [14], and the study of phenophases throughout the year [20].

The foundational era of spiral visualizations established both the mathematical basis for cyclical time and the initial parameters for user interaction. Carlis and Konstan (1998) coined the term “serial periodic data” to refer to data that “has a serial (continuous) dimension that exhibits periodicity,” introducing the Archimedean spiral as a compact alternative to linear plots due to its constant rate of expansion [5]. Frank’s (1998) taxonomy of conceptual models of time similarly supports the notion that cyclic time has attributes distinct from ordinal, interval, and branching models [11]. Building on these concepts, researchers began defining the design space for interacting with these complex structures; Ward and Lipchak (2000) utilized multivariate glyphs to allow users to modify cycle lengths [30], while Weber et al. (2001) expanded the metaphor to support nominal and quantitative data across multiple datasets [31]. Draper et al. (2009) classified these visualizations as “radial space filling” (RSF) patterns, which combine the appearance of pie charts with polar plot techniques [9]. These early systems introduced the first standard UI controls—such as scale, offset, and neighborhood comparison. However, as Kumar et al. (2005) note, despite their space efficacy compared to simple linear plots, scalability remained an issue during this era [17].

The 2010s saw a focus on addressing scalability and enhancing user interaction. Zhao et al. (2011) developed KronoMiner, a tool that utilizes multiple coordinated views—including large-scale overviews and multi-foci temporal hierarchies—to facilitate side-by-side comparison and data abstraction through filtering and zooming [33]. Its MagicAnalytic Lens is a technique that performs zooming and compares parts of time series in real-time, and additional features include tool-tip messages and popup menus [24]. Albers et al. (2014) identified three critical design variables: visual (e.g., color or orientation), mapping, and computational (compression methods) [2]. Their findings suggest that visual variables like color better support pre-attentive summarization, while position offers higher perceptual fidelity for point comparisons. Technological solutions to “spiral explosion” included the fisheye view proposed by Wang (2016), which adjusts focus coefficients and time granularity [29]. Design considerations also extended to cognitive load; for instance, Guo et al. (2017) found that while a 12-hour metaphor is intuitive, a 24-hour clock-face design can be more effective for accuracy and speed [13]. Furthermore, Ceneda et al. (2018) introduced data-driven guidance techniques that provide visual cues to assist users in configuring spiral parameters, such as segments per cycle and band scale, to reveal hidden cyclical patterns [6].

In recent years, research has shifted toward dynamic methods and high-dimensional time-series data. In their survey, Fang et al. (2020) classified these approaches into static and animation methods, accounting for complex characteristics like branch time and multi-angle time [10]. Jiang et al. (2020) further explored dynamic visualizations using video formats [16].

3 FORMATIVE STUDY

To understand the needs of users tracking their sleep, we conducted a formative study focused on young adults in the Boston area. We conducted high-quality, semi-structured interviews with six partic-

ipants to gain a detailed understanding of how they interact with sleep visualizations and identify existing research gaps. The interview guide can be found in Appendix A.

3.0.1 Participant Demographics and Expertise

Our cohort consisted of six participants (5 men, 1 woman) aged 20-35. The occupations included software developer, software engineer, video game developer, PhD student, undergraduate student, and pastoral resident. To ensure diverse perspectives on data presentation, we recruited participants with varying visualization expertise: two beginners, two with intermediate experience, and two with extensive experience.

3.0.2 Device Usage and Tracking Habits

Participants utilized a range of wearable ecosystems: Two users used Fitbit, two used Apple Watch, one used a Galaxy Watch (formerly Fitbit), and one wore both an Apple Watch and a WHOOP strap nightly (having also formerly used an Oura ring before it was lost). Device choice was not data-driven; instead, trackers were typically gifted or chosen because they fit into an existing smartphone ecosystem. Most participants reported nightly use and have been consistent trackers for 1–5+ years. One user supplemented their wearable data with a clinical sleep apnea test involving chest sensors. Participants did not report tracking any data external to their wearable devices, with the exception of one participant who reported recording nightly dreams in a journal.

3.0.3 Motivations and Behaviors

While participants expressed a desire for “active” tracking to improve sleep schedules, most behavior remains passive. Users generally look at whatever visualizations are presented to them rather than actively seeking out specific information. Although a range of sleep goals were reported—from improving duration and quality to gamifying sleep and validating personal experiences of tiredness—many participants resorted to independent personal efforts to adjust their schedules. Consequently, they focus on metrics most relevant to these specific needs, such as duration, wake time, heart rate, and sleep stages.

Participants are also interested in external metrics that are currently perceived as too inconvenient to track, such as food, exercise, caffeine, alcohol, and hydration. This stems from a deep interest in the “why” behind poor sleep; users expressed a strong desire to see causal links between daytime activities—like stress or workouts—and overnight recovery. However, these causal relationships are not readily visualized in current applications. In terms of behavioral change, while users have adjusted sleeping positions, taken melatonin, or used snoring devices over the years, these changes were generally not data-driven or in response to sleep tracking. Instead, they were based on personal experience or external advice.

3.0.4 Visualization and Interface Findings

While current apps provide a wealth of data, participants frequently described much of the interface as uninformative or overwhelming. Many users reported scrolling past notifications and technical graphs—such as respiratory rate, blood oxygen (O2) saturation, and skin temperature—noting that these metrics often lack a clear frame of reference or actionable meaning. Specifically, participants expressed skepticism toward proprietary “Sleep” or “Readiness” scores, citing instances where a “high” score contradicted their actual feelings of exhaustion. One user noted that despite these scores, they still struggled to understand the specific actions needed to feel truly rested.

In terms of effective encodings, the nightly sleep stage timeline (hypnogram) was identified as the most valued and frequently visited visualization. Rather than analyzing long-term historical trends, participants preferred “benchmark charts” that compared the

previous night’s data to their own personal baseline or averages. This preference for “the simpler the better” highlights a gap in current systems; for instance, while users want to adjust their sleep schedules, they find that existing apps do not adapt to their actual lives. Instead of generic reminders, participants suggested that interfaces should provide incremental guidance—such as shifting a sleep alert from 1:00 AM to 12:45 AM—to help them move toward their goals through manageable, data-informed changes.

4 DESIGN AND IMPLEMENTATION

Building on the requirements identified in the formative study and the technical precedents in spiral time-series literature, we developed a cyclical sleep visualization tool designed to address the user desire for actionable, personalized baselines. The system translates standard data from consumer sleep trackers into a spiral encoding to reveal periodic phenomena that are often obscured in conventional views shown to users.

The visualization is built on an Archimedean spiral, chosen for its constant rate of expansion which facilitates compact side-by-side comparison of daily cycles. Specific design features include:

1. **Temporal Navigation and Focus:** To mitigate the “spiral explosion” common in long-term data, we implemented a fish-eye effect that allows users to dynamically zoom into a specific time period while maintaining the broader temporal context. Band size of the rings in the spiral are also adjustable for easier, more focused viewing.
2. **Clock-Face Encoding:** Based on findings regarding cognitive load and accuracy, the visualization utilizes a 24-hour clock-face design with a clear date-change line at midnight to provide a familiar frame of reference.
3. **Multivariate Conditional Layers:** To address the desire for causal links between daytime activity and sleep, users can toggle conditional attributes (ex: caffeine intake or exercise). These layers use a fade-out effect on hover to highlight specific days where these factors were present.
4. **Aggregated Summary View:** A secondary visualization collapses the temporal dimension into a single 24-hour circular band by layering every recorded night of sleep as semi-transparent overlays. This composite view reveals the “shape” of the user’s overall sleep distribution and typical boundaries, serving as a high-density baseline.
5. **Baseline and Benchmarking:** To provide the “normal for me” context that users desire, the system allows users to toggle the aggregated summary view as a background layer. By visualizing sleep on top of this aggregate, users can perform immediate benchmarking to see how their most recent night aligns with or deviates from their long-term patterns.
6. **Environmental Context:** The background integrates sunrise and sunset lines to provide a naturalistic reference for sleep-wake cycles.

A proof-of-concept demonstration of the visualization is published at <https://nathanaelj.github.io/SnoozeSpiral>. The visualization is illustrated in Fig. 1.

5 EVALUATION METHODOLOGY

We propose a within-subjects comparative think-aloud study to validate whether cyclical encoding improves intuition regarding sleep habits compared to existing linear interfaces.

5.0.1 Participants and Data Preparation

We recruited five participants (4 men, 1 woman) aged 20-35, including graduate students and software developers. Participants had varying levels of visualization expertise. Prior to the study, participants shared 30 days of sleep data (stages, duration, and metadata like exercise or caffeine) to populate a personalized version of the cyclical tool.

5.0.2 Experimental Procedure

The study utilized a counterbalanced order to mitigate learning effects (see Appendix B for protocol). For each participant:

- **Linear Condition:** Participants explored their current sleep tracking app, identifying recent trends and personal insights.
- **Cyclical Condition:** Participants interacted with the spiral visualization, utilizing the fisheye zoom and conditional overlays to interpret the same 30-day period.
- **Comparative Interview:** Participants answered side-by-side comparison questions focusing on feature saliency, clarity of patterns, and the actionability of the reflections generated by each view.

5.0.3 Analysis

The study employed descriptive qualitative analysis to evaluate participant feedback. Following the sessions, the authors conducted a collaborative debriefing and review of interview notes and transcripts to identify recurring patterns in user behavior and perception. Primary metrics for success included the participants' ability to uncover new insights, reduced ambiguity in identifying irregularities, and the frequency of actionable reflections.

6 RESULTS

User Experience and Pattern Recognition Overall, participants responded positively to the cyclical visualization, noting its uniqueness and its efficacy in providing a high-level overview of long-term sleep trends. Unlike standard linear representations, the cyclical layout allowed users to quickly identify consistent bedtimes, wake times, and mid-night disturbances. Notably, the design was effective at representing polyphasic sleep. Participants highlighted that the system handled naps with significantly more clarity than industry standards like Apple Health, which users reported "compresses" daily data when naps occur, often rendering the visualization illegible.

Shift in Mental Models: Summary vs. Snapshot The study revealed a distinct shift in how users interacted with the cyclical tool compared to traditional sleep apps. In walkthroughs of their existing setups, participants typically engaged in short-term check-ins (ex: viewing the previous night's duration, daily steps, and calories). Conversely, this tool was viewed as a long-term pattern summary. This shift influenced the frequency of intended use; for instance, one participant noted they would not use the tool for a morning update but would instead consult it weekly to evaluate broader trends.

Feature Preferences and Customization The Personal Average feature stood out to users, though it sparked a debate regarding temporal relevance. While many appreciated the static baseline, one participant suggested a rolling average to better reflect recent behavioral shifts.

User preferences for the temporal interface were split:

- **Clockface Encodings:** Four participants preferred a 12-hour clockface, citing a lack of frequent napping, while one preferred a 24-hour clockface.

- **Geometric Preference:** Participants were evenly split between the traditional and cyclical views. However, the cyclical view elicited strong qualitative feedback regarding the nature of time, with one participant noting that "circles naturally feel 'timier' than bars."

Usability and Interaction Most participants successfully navigated the interactive toolkit, including the zoom, fisheye, and filtering functions. While usability was generally high, some tension emerged regarding control density: a subset of users felt the interface was over-engineered, while others requested even more granular controls, such as the ability to filter by specific sleep stages.

Challenges and Design Friction Despite the general enthusiasm from participants, several barriers to adoption were identified:

- **Data Integrity:** The clarity of visual trends was heavily dependent on data continuity; participants with "patchy" or inconsistent data found the visualizations difficult to interpret.
- **The Tracking Paradox:** Several users expressed a desire for more contextual tags (ex: alcohol and caffeine intake) but remained unwilling to perform the manual logging required to generate that data.
- **Desire for Actionable Insight:** Participants frequently requested more than just descriptive data. There was a strong demand for prescriptive analytics, with users explicitly asking for a one line summary of actionable advice based on their visualizations.
- **External Utility:** One participant identified a potential use case beyond personal tracking—providing the visualization to professional coaches to evaluate the relationship between sleep hygiene and athletic performance.

7 DISCUSSION

Our study demonstrates that while cyclical visualizations successfully reveal long-term sleep trends, reformatting data does not inherently ensure actionability. Despite gaining deeper insights, participants remained burdened by the cognitive task of translation, showing that intuitive visualizations alone cannot bridge the gap between descriptive data and prescriptive behavior change.

These findings are limited by a pro-reflection bias among our participant pool of young professionals and active trackers, whose motivations may not represent more ambivalent users or clinical populations. Furthermore, the proprietary siloing of sleep data and established user habits create significant switching costs, highlighting steep barriers to adoption unless they are deeply integrated into existing personal informatics ecosystems.

8 CONCLUSION

This research demonstrates that while cyclical sleep visualizations effectively address the long-term overview gap in current personal informatics, the transition from descriptive insight to prescriptive action remains a significant challenge. Our formative study highlighted a fundamental disconnect in existing tools, where participants expressed a strong desire for data-informed guidance over static, passive metrics. Our evaluation reveals that while spiral-based encodings improve the recognition of periodic patterns and reveal new insights, the ability to identify a trend does not inherently empower a user to change it. Ultimately, this work highlights that while cyclical metaphors provide a new structural view of sleep, future systems must pair these visualizations with automated analytical layers to translate temporal patterns into concrete health recommendations.

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A SEMI-STRUCTURED INTERVIEW GUIDE

1. Device Context and History

- (a) Which sleep tracking device are you currently using?
- (b) Why did you choose this specific tracker over alternatives?
- (c) Have you used different trackers or systems in the past? If so, why did you switch?

- (d) How long have you been tracking, and what is your typical frequency of usage?

2. Motivations and Goals

- (a) What was your primary motivation for starting to track your sleep?
- (b) Would you describe your tracking as more passive (checking data out of curiosity) or goal-oriented (trying to achieve a specific outcome)?
- (c) What specific sleep goals are you currently focused on? (e.g., increasing duration, improving quality, adjusting schedule).
- (d) Since you first started tracking, have you changed any habits or behaviors?

3. External Factors and Limitations

- (a) Do you track any factors related to your sleep outside of your primary app?
- (b) Is there anything you want to track but find too inconvenient to log manually?

4. Interface and Visualization Evaluation

- (a) Which features or metrics do you look at most frequently?
- (b) What information in your app do you find uninformative? Are there specific charts or notifications that feel like "visual clutter" or get in the way?
- (c) Is there anything that you wish you could visualize but are currently unable to?

B EVALUATION INTERVIEW PROTOCOL

Because the study aimed to capture the participants' unfiltered intuition regarding cyclical versus traditional views of their sleep, the interviews followed an unstructured, open-ended format. The primary prompt used for both conditions was:

"Please walk me through what you notice about your sleep habits while looking at this visualization. Feel free to point out any patterns, anomalies, or insights that stand out to you."

To ensure a consistent depth of reflection across all five participants, the authors used neutral probes when natural pauses occurred. For instance, "What makes that specific part stand out to you?" or "How does this compare to how you usually view your sleep data?"

The session concluded with a semi-structured retrospective comparison. The researchers prompted participants to elaborate on specific differences or similarities they had noted during the individual walk-throughs, allowing for a deeper reflection on their preferences and the perceived utility of each encoding.